

Paper D-13: Population Genetics: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD

Independent Researcher

Founder, Vura Labs

Kampala, Uganda

ORCID: 0009-0001-0556-5516

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Abstract

This paper develops a falsifiable CDFD/AFL model of Population Genetics. It maps mutation, migration, selection, recombination, and reproduction to driving flux Φ , population size, connectivity, resources, drift, and reproductive limits to active constraint C , recombination, plasticity, dispersal, mate choice, and adaptation to responsiveness S , and allele frequencies, bottlenecks, linkage, environment, and selection history to structural memory M_s . The target outcome is diversity, adaptation, fixation, and extinction risk, tested against genomes, pedigrees, experimental evolution, demography, and environmental records. The paper treats universality as a shared model architecture rather than an assertion that unlike systems have the same material mechanism. It states the negative result that would make the mapping unnecessary and places the domain claim inside the cumulative CDFD programme and its reproducible runtime boundary.

Symbol Conventions

CDFD/AFL Part IV symbol convention.

Symbol	Part IV meaning	Cross-domain reading
Φ	Driving flux or throughput	Energy, matter, signal, capital, information, traffic, force, or biological load entering a system
C	Active constraint or capacity burden	Bottleneck, resistance, boundary, scarcity, toxicity, regulation, topology, latency, or load-bearing limit
S	Responsiveness or adaptive surface	The system's ability to reconfigure, buffer, route, repair, learn, or preserve function under changing load
M_s	Structural memory	Stored damage, hysteresis, inherited topology, institutional memory, trained weights, ecological legacy, or retained state
Ψ_s	Adaptive operating ratio $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation, overload, recovery, or regime change; not a universal constant
Λ	Life Number or capstone surplus ratio where explicitly defined	Reserved for Part II-derived life-number notation or synthesis-level surplus measures, never an undefined decoration

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Accumulation or reinforcement coefficient	Sets how fast a pathway, constraint, habit, cascade, or retained structure grows under load
β	Relaxation, decay, clearance, or washout coefficient	Sets loss, recovery, damping, forgetting, degradation, or return toward baseline
γ	Coupling, diffusion, redistribution, or transfer coefficient	Sets spread across nodes, layers, tissues, markets, infrastructures, media, or physical phases
δ, ϵ	Perturbation, tolerance, small offset, or numerical guard	Used only as local thresholds, stability margins, measurement tolerances, or stress increments
η, λ, μ, ν	Efficiency, rate, baseline, intensity, or frequency terms	Meaning is fixed by the equation, subscript, and domain-specific proxy in the paragraph where the term appears
ρ, σ, τ, ϕ	Density/correlation, variability/noise, time-scale, phase, or field terms	Local coefficients only; subscripts bind them to the named pathway, domain, or experiment
Ω, Λ	Domain/state space; Life Number where defined	Ω names a modeled state space; Λ carries life-number or synthesis notation only when the paper defines it

1 Series Context

Part I sets the flow-constraint language used here [3]. Part II develops the Life Number and the origins-of-life use of the same notation [4]. Part III applies AFL to biology and medicine

[5]. Part IV [6], DOI <https://doi.org/10.5281/zenodo.20395901>, is the cross-domain stress test: it asks where the notation clarifies measurement, and where it becomes only a metaphor.

2 Introduction

2.1 The Muggabi Universal Capacity Law

Building directly upon the Fundamental Physics (Part I) [3], the **Muggabi Universal Capacity Law** proposes that across all scales—from black hole event horizons to global financial markets and power grids—driving high flux (Φ) against a rigid topological constraint (C) can drive system saturation. When Φ/C approaches critical thresholds, the system does not scale linearly; it undergoes a topological phase transition.

2.2 The Muggabi Universal Memory Law

The **Muggabi Universal Memory Law** frames persistent systemic collapse. When a complex network fails to clear its structural memory (M_s) and its relaxation parameter decays ($\tau_{relax} \rightarrow \infty$), temporary stressors become permanent rigidities. This is the proposed CDFD mechanism behind ecological death, economic depressions, and infrastructural gridlock.

Evolution is the algorithm by which life explores morphological and functional space. Mutations introduce random variations, gene flow mixes them across geography, and natural selection eliminates variants that are ill-suited to the environment.

In CDFD, this process is formally comparable at the declared-variable level to fluid dynamics. The genome of a population is a fluid network of information passing through time. The environment acts as a restrictive channel that shapes and squeezes this flow.

3 The Gene Pool as an Adaptive Surface

We track the evolutionary stability of a species via the Universal Adaptive Ratio:

$$\Psi_s = \left(\frac{\Phi}{C} \right) \cdot S \cdot M_s \quad (1)$$

In the context of population genetics:

- **Flux** (Φ): The rate of genetic variation introduced by mutation, recombination, and immigration (gene flow). This is the raw informational throughput of the species.
- **Constraint** (C): Natural selection and environmental pressure (predators, climate, resource scarcity). Selection acts as a filter, removing non-viable informational flow.
- **Responsiveness** (S): The genetic diversity and phenotypic plasticity of the population. A highly diverse population can easily flex to meet changing constraints.
- **Memory** (M_s): Genetic fixation and canalization. As a trait becomes heavily selected for, it becomes fixed in the genome. The species "remembers" successful past adaptations, severely restricting future morphological deviations (high M_s).

3.1 Extinction vs. Speciation

A well-adapted species in a stable environment operates at $\Psi_s \approx 1$. The genetic flux (Φ) perfectly matches the environmental constraints (C), and the genetic memory (M_s) maintains the optimal phenotype.

If the environment suddenly changes (e.g., rapid climate change), the constraint C spikes violently. If the species has low diversity (S is low) and its genome is heavily canalized (M_s is locked), it cannot adapt. The ratio plummets to $\Psi_s \ll 1$. The flow of genes through time ceases entirely—this is **Extinction**.

Conversely, consider a population spread across a vast area that is suddenly divided by a mountain range. The gene flow (Φ) between the two groups drops to zero. Isolated, each group faces unique local constraints (C). To restore $\Psi_s \approx 1$ locally, the responsiveness (S) of each group explores different mutational pathways. Over time, their genetic memories (M_s) lock into incompatible configurations. The singular adaptive surface has bifurcated into two distinct species. This is **Speciation**, a direct topological consequence of severing the flux network.

4 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system’s ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

4.1 Current Runtime Diagnostic: Universal Network Cascade

The release-local discovery script keeps this paper tied to the current CDFD Runtime [7], including RK4 field updates, surface dynamics, structural-memory diffusion, and a finite-value audit. In the 2026-06-04 runtime pass, the localized hub-drive stress test ended with hub $\Psi_s = 5.21$, peak network $\Psi_s = 5.21$, 21 nodes above $\Psi_s > 1.5$, and 37 maximum memory-locked nodes. I treat those numbers as a runtime sanity check and as a target for later domain measurement, not as a substitute for field data.

5 Major Universal Upgrade: Mechanism, Scale, and Tests

5.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Population Genetics, the proposed drive is mutation, migration, selection, recombination, and reproduction. The active limitation is population size, connectivity, resources, drift, and reproductive limits. Responsiveness is carried by recombination, plasticity, dispersal, mate choice, and adaptation, while retained state is represented by allele frequencies, bottlenecks,

linkage, environment, and selection history. The outcome to explain is diversity, adaptation, fixation, and extinction risk.

Table 1: Operational CDFD/AFL map for Paper D-13.

Term	Paper-specific observable meaning	Measurement question
Φ	mutation, migration, selection, recombination, and reproduction	What rate, load, gradient, or demand enters the focal system?
C	population size, connectivity, resources, drift, and reproductive limits	Which measured bottleneck limits transfer, function, or recovery?
S	recombination, plasticity, dispersal, mate choice, and adaptation	Which process changes effective capacity or reroutes the load?
M_s	allele frequencies, bottlenecks, linkage, environment, and selection history	Which retained state makes equal present loads produce different futures?
Y	diversity, adaptation, fixation, and extinction risk	Which outcome is predicted out of sample, with uncertainty?

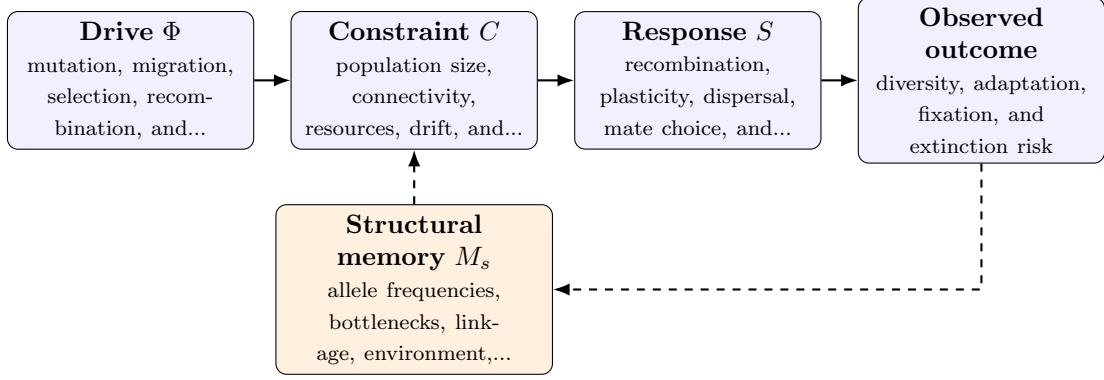


Figure 1: Paper D-13 observable CDFD/AFL architecture for Population Genetics. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

5.2 Minimal Coupled Model

A paper-level model must separate throughput, constraint accumulation, response, and history. One deliberately minimal form is

$$Y(t) = \frac{\Phi(t)S(t)}{\epsilon + C(t)}, \quad (2)$$

$$\frac{dC}{dt} = \alpha\Phi(t) - \beta S(t)C(t) + \gamma\mathcal{L}C(t), \quad (3)$$

$$\frac{dM_s}{dt} = \eta C(t) - \mu M_s(t), \quad (4)$$

$$\Psi_s(t) = \frac{\hat{\Phi}(t)}{\epsilon + \hat{C}(t)} \hat{S}(t) [1 + \omega \hat{M}_s(t)]. \quad (5)$$

Here Y is the measured output, $\epsilon > 0$ prevents a singular denominator, $\mathcal{L}$ is a spatial or network coupling operator, and hats denote explicitly normalized variables. The sign and magnitude of ω must be estimated: memory can preserve useful organization or amplify damage. No fixed threshold is universal before the variables, normalization, and observation scale are declared.

For this paper, the hypothesized causal sequence is specific. A change in mutation, migration, selection, recombination, and reproduction arrives faster than recombination, plasticity, dispersal, mate choice, and adaptation can compensate. This raises or redistributes population size, connectivity, resources, drift, and reproductive limits. If the load persists, the retained state encoded by allele frequencies, bottlenecks, linkage, environment, and selection history changes the next response, so a later event of equal magnitude can produce a different trajectory in diversity, adaptation, fixation, and extinction risk. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

5.3 Cross-Scale Universality Without Mechanistic Erasure

For the domain applications family, the universal comparison is a disciplined translation test across systems with very different material mechanisms. A valid mapping preserves the baseline science of the domain, declares the scale of every variable, and earns its place through a discriminating prediction rather than resemblance alone. [2, 9, 8, 1] The CDFD programme is cumulative: Part I supplies the flow-constraint foundation [3]; Part II asks when maintained throughput supports persistent organized states [4]; Part III develops adaptive limitation and biological memory [5]; and Part IV [6] tests how much of that architecture survives translation. The CDFD Runtime [7] supplies a reproducible numerical stress surface, but it does not substitute for the domain evidence named here.

The required scale declaration for this paper is generations to geological time; loci to metapopulations. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

5.4 Evidence Ladder and Discriminating Tests

The strongest practical design combines a prospective perturbation with a recovery window. The load-ramp phase estimates where constraint begins to rise faster than output. The recovery phase estimates β and μ , separating fast relaxation from persistent memory. A topology or coupling intervention then asks whether rerouting changes the cascade as predicted. Null results are informative because they identify domains in which the universal notation adds no measurable value.

5.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from genomes, pedigrees, experimental evolution, demography, and environmental records and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of diversity, adaptation, fixation, and extinction risk. Fourth,

Table 2: Evidence ladder for Paper D-13. Each rung can fail independently.

Test	Design	Discriminating result
Proxy audit	Measure genomes, pedigrees, experimental evolution, demography, and environmental records and preregister how each record maps to Φ , C , S , M_s , and Y .	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe a selection or migration shift across populations with different histories while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper’s stated falsifier is: explicit memory and constraint variables do not improve evolutionary prediction.

repeat the analysis across the declared scale range and publish negative cases as part of the universality audit.

Three boundaries are non-negotiable. Correlation between flow and failure does not identify a constraint mechanism. A fitted memory term does not prove that the stored state has the same material meaning in another domain. Finally, a runtime cascade is evidence about the implementation, not evidence that the real system follows it. The paper becomes stronger when it can say precisely where the translation stops.

6 Conclusion

The mathematics of evolution are the mathematics of constrained flow. By applying CDFD to population genetics, we eliminate the false dichotomy between biology and physics. The gene pool flows, adapts, and breaks around environmental constraints in a comparable but materially distinct manner that a river carves a canyon or a language splinters into dialects.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Part-specific runtime diagnostics are under each Part’s `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

7 Reference Layer

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